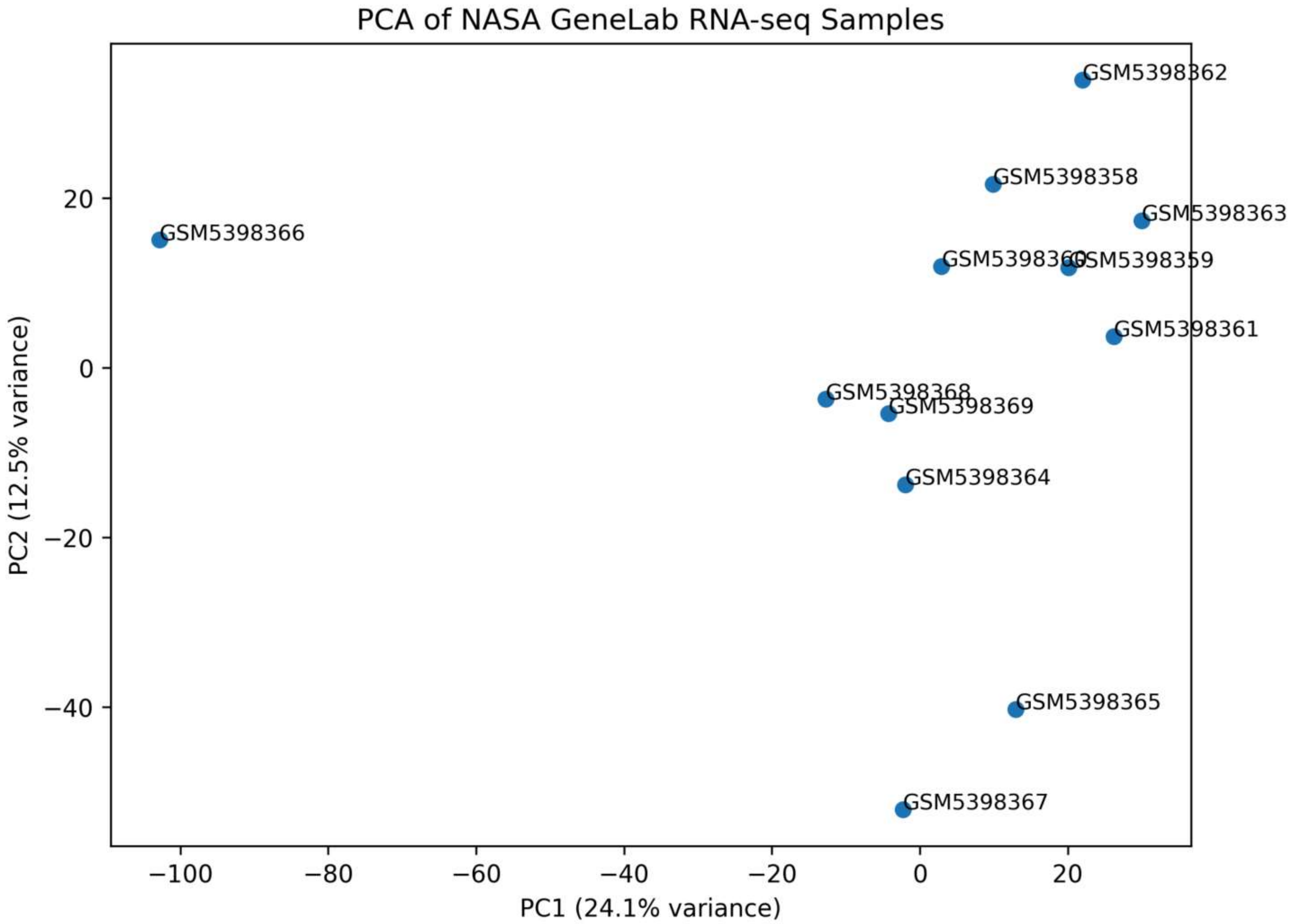
🚀 NASA RNA-seq Bioinformatics Dashboard

Differential expression analysis of NASA GeneLab RNA-seq samples

Samples	Upregulated Genes	Downregulated Genes
12	135	670

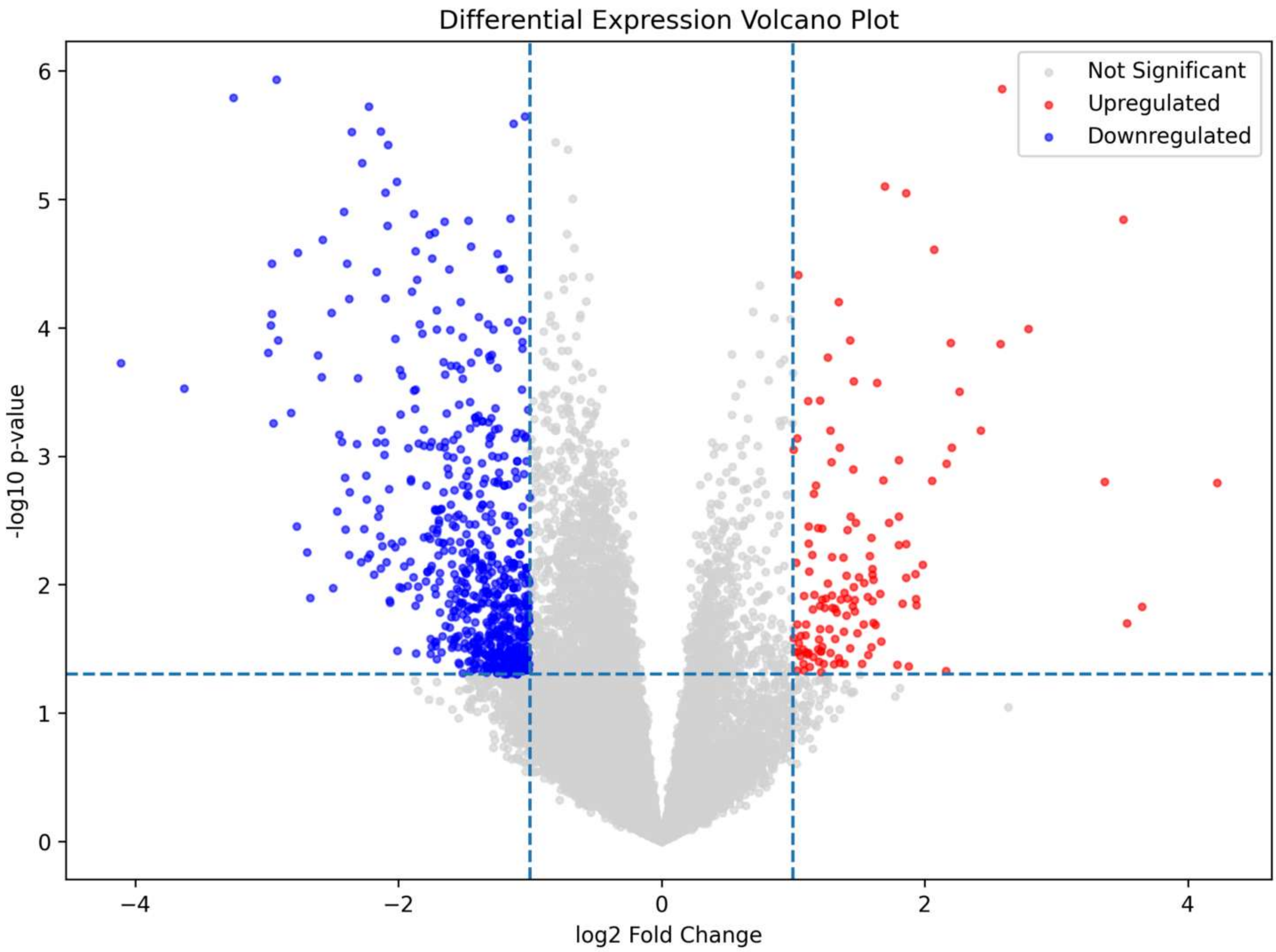
🧬 PCA Analysis



Principal Component Analysis of RNA-seq samples

Principal Component Analysis (PCA) shows clustering of transcriptomic profiles between experimental groups.

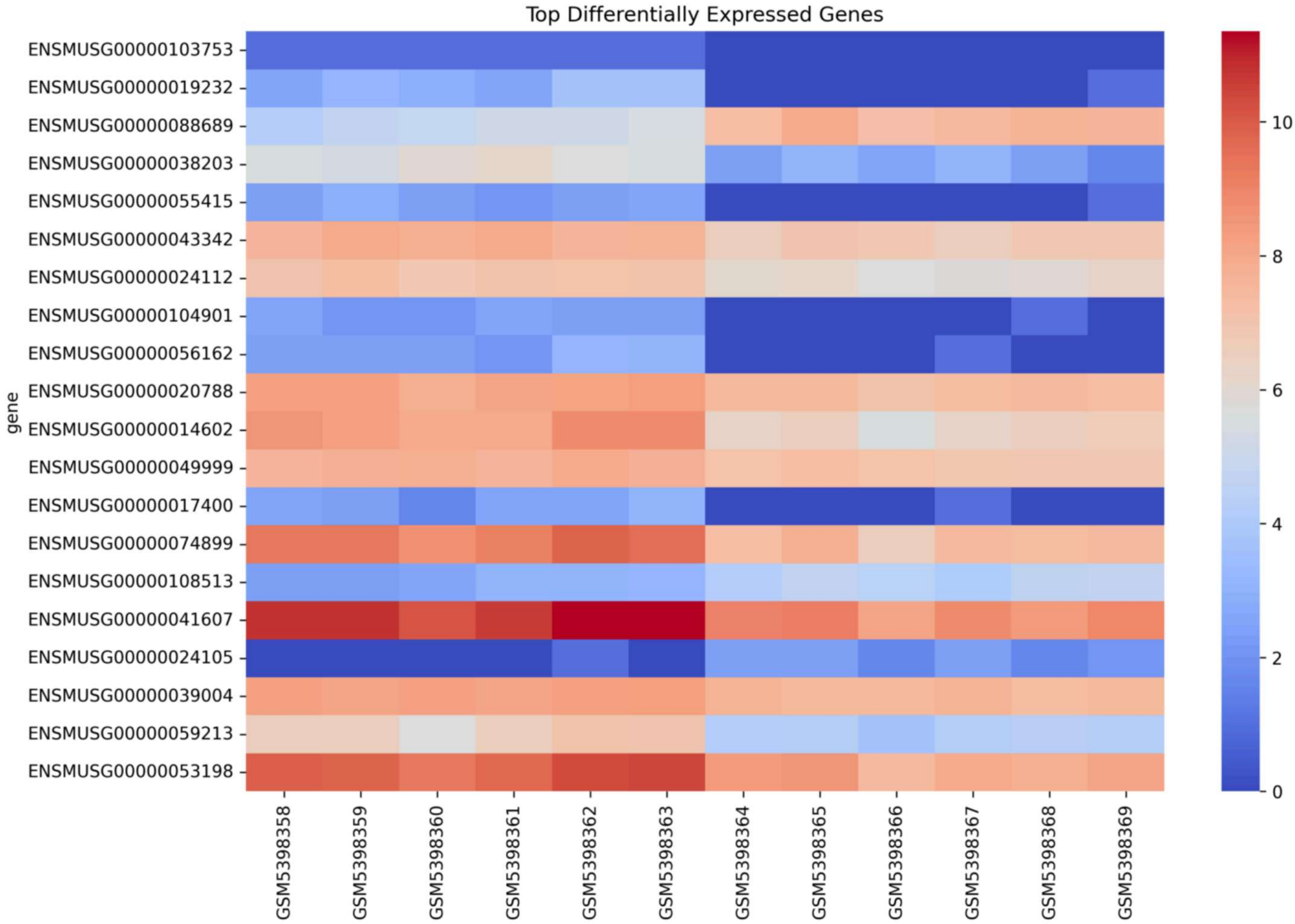
🌋 Differential Expression Volcano Plot



Differentially expressed genes

- 🔴 Red points = Upregulated genes
- 🔵 Blue points = Downregulated genes
- 🟣 Gray points = Not significant

🔥 Heatmap of Top Differentially Expressed Genes



Expression patterns across samples

📊 Top Differentially Expressed Genes

	gene	log2FC	pvalue	neg_log10_pvalue
0	ENSMUSG00000103753	-1	0	
1	ENSMUSG00000019232	-2.9247	0.000001	
2	ENSMUSG00000088689	2.5853	0.000001	
3	ENSMUSG00000038203	-3.2546	0.000002	
4	ENSMUSG00000055415	-2.2264	0.000002	
5	ENSMUSG00000043342	-1.0388	0.000002	
6	ENSMUSG00000024112	-1.1249	0.000003	
7	ENSMUSG00000104901	-2.1356	0.000003	
8	ENSMUSG00000056162	-2.356	0.000003	
9	ENSMUSG00000020788	-0.8082	0.000004	

🛠 Technologies Used

- Python
- Pandas
- NumPy
- Scikit-learn
- Matplotlib

- Seaborn
- Streamlit
- Docker

📡 Bioinformatics Workflow

RNA-seq → normalization → PCA → differential expression → volcano plot → heatmap